

Practical No- 4

Aim :- To develop a Customer Order Management System & retrieve data using SQL joins (INNER, LEFT, RIGHT, FULL OUTER) to understand relationships between multiple tables.

Theory :-

In database management system, joins are used to combine rows from two or more tables based on related column between them.

- 1) INNER JOIN :- Returns only the rows that have matching values in both tables.
- 2) LEFT JOIN (or LEFT OUTER JOIN) :- Returns all rows from the left table & matched rows from the right table. If there is no match, the result is NULL on the right side.
- 3) RIGHT JOIN (or RIGHT OUTER JOIN) :- Returns all rows from the right table & matched rows from the left table. If there is no match, the result is NULL on the left side.
- 4) FULL OUTER JOIN :- Returns all rows when there is a match in one of the tables. Unmatched rows from either table appear with NULL in the columns of the other table.

Tables Used :-

- 1) Customers :- Contains customer details.
- 2) Orders :- Contains order details linked to customers via Customer ID.

Conclusion :-

- 1) INNER JOIN :- Showed only Customer who have placed orders.
- 2) LEFT JOIN :- displayed all Customers, including those without orders.
- 3) RIGHT JOIN :- displayed all orders, even those placed by Customers not present in Customers table.
- 4) FULL OUTER JOIN :- provided a complete view of both tables, including unmatched rows.